

Matthew Au-Yeung

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science, Honours Co-op | GPA 3.7

Sept 2022 – Apr 2027

Course: DSA, Operating Systems, Distributed Systems, Concurrency, Compilers, Cryptography & Security

SKILLS

Languages: Java, Kotlin, Python, Rust, Go, JavaScript, TypeScript, C#, Elixir, C/C++

Tools and Frameworks: Spring, gRPC, ASP.NET Core, React, Gin, Auth0, Nix

Databases and Cloud: PostgreSQL, Microsoft SQL Server (MSSQL), Firebase, Docker, Github Actions

EXPERIENCE

Software Engineer Intern

May 2025 – Aug 2025

Mechanical Orchard

Remote

- Implemented a **Kotlin API** for Java-style COBOL objects integrating with a **Rust** core, using lazy allocation and pointer reuse to deliver **40% memory savings** and **10x throughput** in modernized services
- Cut migration time by **60%** by building a **Python MCP** that converts COBOL into Java context for LLMs
- Migrated **2 client mainframe programs** serving inventory for **1K+ stores** using **Java, Spring**, and **gRPC**
- Published an open-source Python API (bit.ly/4f9calf) with **4K downloads** to simplify generating Java files

Software Developer Intern

Jan 2025 – Apr 2025

Ford Motor Company

Waterloo, ON

- Built a **Kotlin Android service** capturing Apple CarPlay button telemetry for **5 vehicle-refinement teams**
- Synchronized **4 background services** with **Coroutines** and **Channels** to monitor CarPlay transitions
- Increased mobile projection test coverage from **70% to 83%** using **Coroutines, JUnit**, and **Robolectric**

Compilers Testing Research Assistant

May 2024 – Jan 2025

University of Waterloo

Waterloo, ON

- Engineered a MIPS assembly emulator and graphical runtime debugger in **Kotlin** used annually by **over 1,000 students** to debug compiler outputs (bit.ly/4gfQ7sh)
- Created an **ANTLR** grammar and parser to collapse bracket-nesting code and minify files from **5 languages**

Software Developer Intern

May 2024 – Aug 2024

Township of Centre Wellington

Fergus, ON

- Shipped a document portal tracking **1K+ locations** using **C# ASP.NET Core, TypeScript**, and **MSSQL**
- Migrated **Auth0** authorization to middleware-enforced policies, reducing external reliance for **28 municipalities**
- Improved communication for **100+ officials** by building shared notes using **SignalR Websockets** and **Quill.js**

Data Science Intern

May 2023 – Aug 2023

Bio-Ag Consultants and Distributors

Wellesley, ON

- Implemented Python Qt apps to log company metrics, sales, and inventory from **Dynamics365** to Excel, saving **over 3 hours daily** of manual data logging

PROJECTS

Distributed Go Channels | Go, gRPC, hashicorp/raft

[m4tth3/dchan](https://github.com/m4tth3/dchan)

- Created a **Go** library providing **native channels** for many-to-many distributed messaging over **Raft** and **gRPC**
- Reduced message latency by building a **gRPC-based Raft Transport** that shares long-lived connections

Timestamp | Android, KMP, Spring Boot, Jetpack Compose, PostgreSQL

[m4tth3/timestamp](https://github.com/m4tth3/timestamp)

- Implemented a meetup app serving realtime ETAs using **Jetpack Compose, Kotlin Multiplatform**, and **Ktor**
- Built a **Spring Boot** backend server integrating **Graphhopper's** routing engine to stream ETAs in **<100ms**